

Elizabeth Muthoni Julius

bethjulius08@gmail.com

+33678948839 | +254797565461

<https://www.linkedin.com/in/elizabeth-julius-2b27ab347>

PROFILE

Master's student in Computer Science specializing in backend development (Python, FastAPI, MERN) and applied AI. Seeking a software engineering internship. Interested in building secure APIs and scalable systems, with hands-on experience in real-world projects, freelancing, and machine learning.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C++

Frameworks: FastAPI, Django, MERN Stack

Tools: Git, Postman, Linux, MySQL

AI / Data: Scikit-learn, Pandas, NumPy, Data Preprocessing, Model Evaluation

Concepts: REST APIs, Authentication, Application Security

PROFESSIONAL EXPERIENCE

Freelance Backend Developer — Web & Backend Projects

Jun 2024 – Present

- Designed and developed web solutions for multiple clients
- Built a residence management system (user and data management)
- Developed a crypto-related web platform (backend architecture and API integration)
- Created a declutter/organization platform for managing personal items
- Applied best practices in API design, security, and system scalability

IT Support, Mentor — Power Learn Project Academy

Jul 2025 – Present

- Mentored students in software project development

- Promoted secure coding practices and technical best practices
- Offered technical assistance to learners.

IT Intern — Tana Water Works Development Agency

Feb 2025 – Apr 2025

- Supported ICT infrastructure and network systems
 - Contributed to ERP system improvements with a focus on data security
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PROJECTS

CreditCard Detection Model

- Built a machine learning-based credit card fraud detection system using XGBoost
- Performed data preprocessing, feature scaling, and model evaluation
- Deployed a FastAPI REST API for real-time fraud prediction and risk classification

Diabetes Prediction Model

- Built a machine learning model to predict diabetes risk
- Performed data preprocessing and evaluated model performance (accuracy, precision, recall)

UAMAS – University Assessment Automation System

- Developed backend services to automate grading processes
- Integrated GPT API and OCR for processing submissions

KaziLink – USSD Job Platform

- Built backend using FastAPI and integrated Africa's Talking API
- Designed scalable and secure system architecture

ElimuTuts – MERN Education Platform

- Developed dashboards and secure authentication system
 - Collaborated using GitHub
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EDUCATION

CESI — Master's in Computer Science (Exchange Program)
Feb 2026 – Present

- IoT, AI & Data Science, Research Projects

Dedan Kimathi University of Technology

Bachelor's in Computer Science
2022 – 2025

- Algorithms, Machine Learning, Networking, Web Development

CERTIFICATIONS

CCNA 1 Networking | PortSwigger Web Security | Postman API Fundamentals | CyberTalents
Linux Basics | Power Learn Project

LANGUAGES

English (Fluent)
Swahili (Fluent)
French (Beginner)